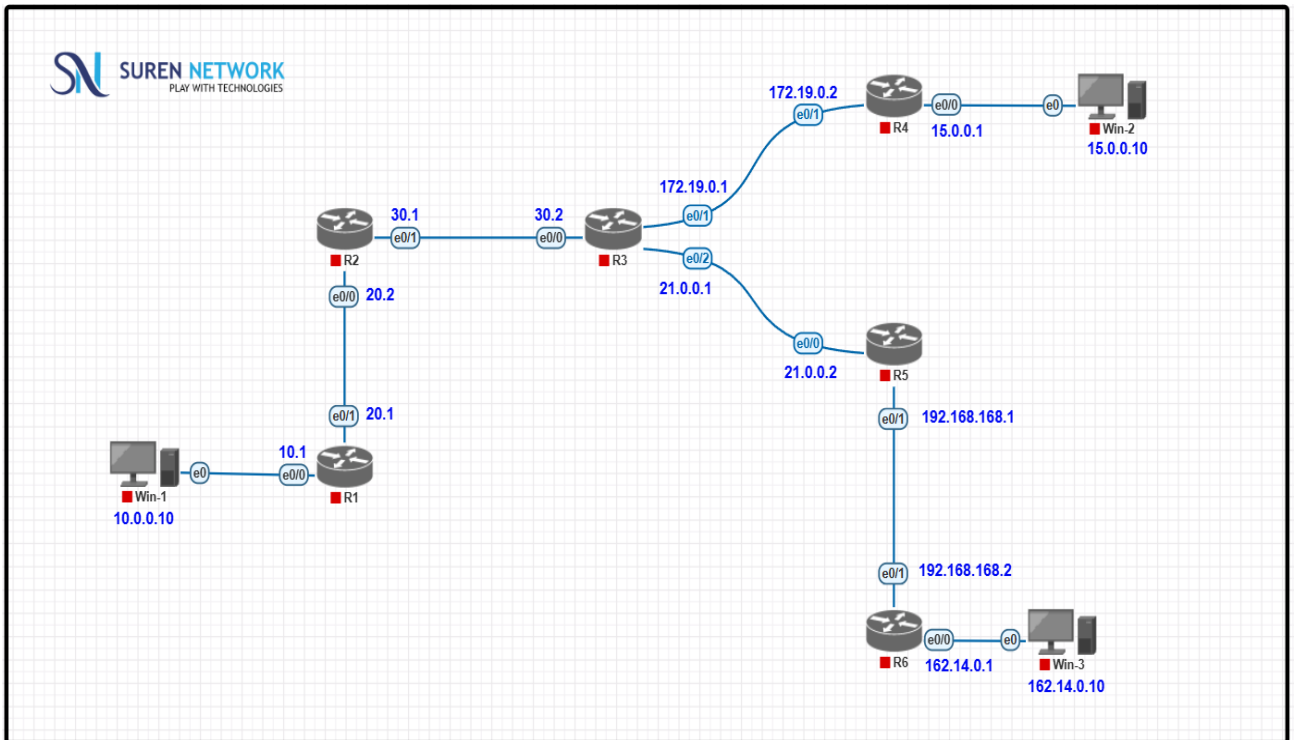


Static Routing Project 1



Router Configuration & Static Routing

```
Enter configuration commands, one per line. End with CNTL/Z.
R6(config)#ip route 10.0.0.0 255.0.0.0 192.168.168.1
R6(config)#ip route 20.0.0.0 255.0.0.0 192.168.168.1
R6(config)#ip route 30.0.0.0 255.0.0.0 192.168.168.1
R6(config)#ip route 15.0.0.0 255.0.0.0 192.168.168.1
R6(config)#ip route 21.0.0.0 255.0.0.0 192.168.168.1
R6(config)#ip route 172.19.0.0 255.255.0.0 192.168.168.1
R6(config)#
R6#
*Jul 18 16:20:59.222: %SYS-5-CONFIG_I: Configured from console by console
R6#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - OMR, P - periodic downloaded static route, R - RIPv2, 1 - LISP
a - application route
* - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

S 10.0.0.0/8 [1/0] via 192.168.168.1
S 15.0.0.0/8 [1/0] via 192.168.168.1
S 20.0.0.0/8 [1/0] via 192.168.168.1
S 30.0.0.0/8 [1/0] via 192.168.168.1
C 162.14.0.0/16 is directly connected, Ethernet0/0
L 162.14.0.1/32 is directly connected, Ethernet0/0
S 172.19.0.0/16 [1/0] via 192.168.168.1
S 192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.168.0/24 is directly connected, Ethernet0/1
C 192.168.168.2/32 is directly connected, Ethernet0/1

R6#sh ip int brief
Interface IP-Address OK? Method Status Prot
Ethernet0/0 162.14.0.1 YES manual up up
Ethernet0/1 192.168.168.2 YES manual up up
Ethernet0/2 unassigned YES TFTP down down
Ethernet0/3 unassigned YES TFTP down down

R6#
```

```
Enter configuration commands, one per line. End with CNTL/Z.
R5(config)#ip route 162.14.0.0 255.255.0.0 192.168.168.2
R5(config)#ip route 10.0.0.0 255.0.0.0 21.0.0.1
R5(config)#ip route 20.0.0.0 255.0.0.0 21.0.0.1
R5(config)#ip route 30.0.0.0 255.0.0.0 21.0.0.1
R5(config)#ip route 15.0.0.0 255.0.0.0 21.0.0.1
R5(config)#ip route 172.19.0.0 255.255.0.0 21.0.0.1
R5(config)#
R5#
*Jul 18 16:14:52.947: %SYS-5-CONFIG_I: Configured from console by console
R5#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - OMR, P - periodic downloaded static route, R - RIPv2, 1 - LISP
a - application route
* - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

S 10.0.0.0/8 [1/0] via 21.0.0.1
S 15.0.0.0/8 [1/0] via 21.0.0.1
S 20.0.0.0/8 [1/0] via 21.0.0.1
C 21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L 21.0.0.0/8 is directly connected, Ethernet0/0
L 21.0.0.2/32 is directly connected, Ethernet0/0
S 30.0.0.0/8 [1/0] via 21.0.0.1
S 162.14.0.0/16 [1/0] via 192.168.168.2
S 172.19.0.0/16 [1/0] via 21.0.0.1
S 192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.168.0/24 is directly connected, Ethernet0/1
C 192.168.168.1/32 is directly connected, Ethernet0/1

R5#sh ip int brief
Interface IP-Address OK? Method Status Prot
Ethernet0/0 21.0.0.2 YES manual up up
Ethernet0/1 192.168.168.1 YES manual up up
Ethernet0/2 unassigned YES TFTP down down
Ethernet0/3 unassigned YES TFTP down down

R5#
```

```
R3#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - OMR, P - periodic downloaded static route, R - RIPv2, 1 - LISP
a - application route
* - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

S 10.0.0.0/8 [1/0] via 30.0.0.1
S 15.0.0.0/8 [1/0] via 172.19.0.2
S 20.0.0.0/8 [1/0] via 30.0.0.1
C 21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L 21.0.0.0/8 is directly connected, Ethernet0/2
L 21.0.0.1/32 is directly connected, Ethernet0/2
C 30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 30.0.0.0/8 is directly connected, Ethernet0/0
L 30.0.0.2/32 is directly connected, Ethernet0/0
C 162.14.0.0/16 [1/0] via 21.0.0.2
S 172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C 172.19.0.0/16 is directly connected, Ethernet0/1

R3#sh ip int brief
Interface IP-Address OK? Method Status Prot
Ethernet0/0 30.0.0.2 YES manual up up
Ethernet0/1 172.19.0.1 YES manual up up
Ethernet0/2 21.0.0.1 YES manual up up
Ethernet0/3 unassigned YES TFTP down down

R3#
```

```
R4
R4(config)#ip route 10.
% incomplete command.

R4(config)#ip route 10.0.0.0 255.0.0.0 172.19.0.1
R4(config)#ip route 20.0.0.0 255.0.0.0 172.19.0.1
R4(config)#ip route 30.0.0.0 255.0.0.0 172.19.0.1
R4(config)#ip route 21.0.0.0 255.0.0.0 172.19.0.1
R4(config)#ip route 162.14.0.0 255.255.0.0 172.19.0.1
R4(config)#ip route 192.168.168.0 255.255.255.0 172.19.0.1
R4(config)#
R4#
*Jul 18 16:07:36.097: %SYS-5-CONFIG_I: Configured from console by console
R4#sh ip int brief
Interface IP-Address OK? Method Status Prot
Gig0/0
Ethernet0/0 15.0.0.1 YES manual up up
Ethernet0/1 172.19.0.2 YES manual up up
Ethernet0/2 unassigned YES TFTP down down
Ethernet0/3 unassigned YES TFTP down down

R4#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, I - ISIS
a - application route
* - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

R
  10.0.0.0/8 [1/0] via 172.19.0.1
C
  15.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C
  15.0.0.0/8 is directly connected, Ethernet0/0
C
  15.0.0.1/32 is directly connected, Ethernet0/0
C
  20.0.0.0/8 [1/0] via 172.19.0.1
C
  21.0.0.0/8 [1/0] via 172.19.0.1
C
  30.0.0.0/8 [1/0] via 172.19.0.1
C
  162.14.0.0/16 [1/0] via 172.19.0.1
C
  172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C
  172.19.0.0/16 is directly connected, Ethernet0/1
C
  172.19.0.2/32 is directly connected, Ethernet0/1
C
  192.168.168.0/24 [1/0] via 172.19.0.1
R4#
```

```
R2
R2(config)#
R2(config)#ip route 162.14.0.0 255.255.0.0 30.0.0.2
R2(config)#
R2#
*Jul 18 15:47:56.095: %SYS-5-CONFIG_I: Configured from console by console
R2#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, I - ISIS
a - application route
* - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

R
  10.0.0.0/8 [1/0] via 20.0.0.1
C
  15.0.0.0/8 [1/0] via 30.0.0.2
C
  20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C
  20.0.0.0/8 is directly connected, Ethernet0/0
C
  20.0.0.2/32 is directly connected, Ethernet0/0
C
  21.0.0.0/8 [1/0] via 30.0.0.2
C
  30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C
  30.0.0.0/8 is directly connected, Ethernet0/1
C
  30.0.0.1/32 is directly connected, Ethernet0/1
C
  162.14.0.0/16 [1/0] via 30.0.0.2
C
  172.19.0.0/16 [1/0] via 30.0.0.2
C
  192.168.168.0/24 [1/0] via 30.0.0.2
R2#
R2#sh ip int brief
Interface IP-Address OK? Method Status Protocol
Gig0/0 20.0.0.2 YES manual up up
Ethernet0/1 30.0.0.1 YES manual up up
Ethernet0/2 unassigned YES TFTP down down
Ethernet0/3 unassigned YES TFTP down down
R2#
```

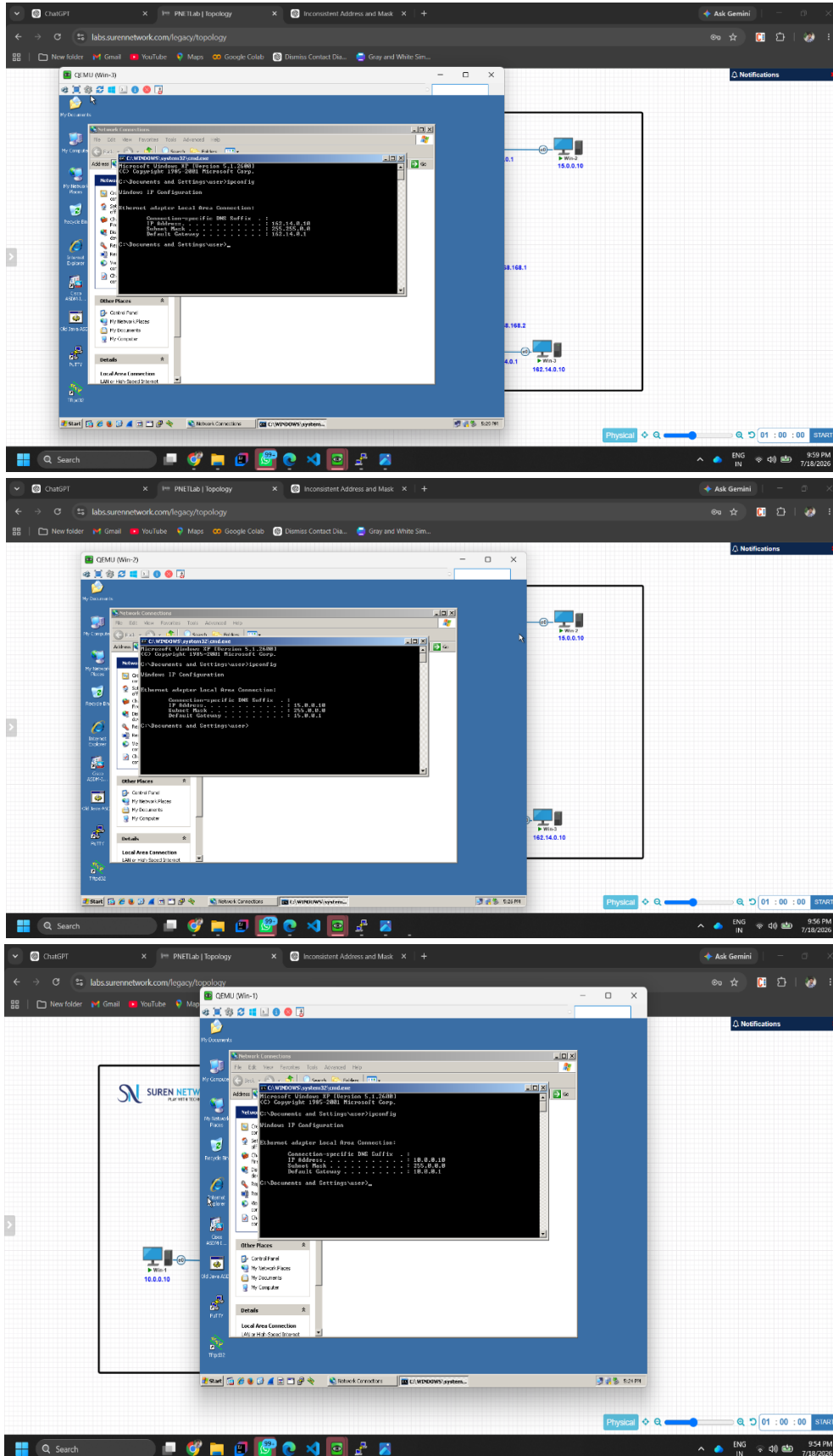
```
R1
R1(config)#ip route 172.19.0.0 255.255.0.0 20.0.0.2
R1(config)#ip route 15.0.0.0 255.0.0.0 20.0.0.2
R1#
% Invalid input detected at '^' marker.

R1(config)#ip route 15.0.0.0 255.0.0.0 20.0.0.2
R1(config)#ip route 21.0.0.0 255.0.0.0 20.0.0.2
R1(config)#ip route 192.168.168.0 255.255.255.0 20.0.0.2
R1(config)#ip route 162.14.0.0 255.255.0.0 20.0.0.2
R1(config)#
R1#
*Jul 18 15:29:45.589: %SYS-5-CONFIG_I: Configured from console by console
R1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, I - ISIS
a - application route
* - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

R
  10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C
  10.0.0.0/8 is directly connected, Ethernet0/0
C
  10.0.0.1/32 is directly connected, Ethernet0/0
C
  15.0.0.0/8 [1/0] via 20.0.0.2
C
  20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C
  20.0.0.0/8 is directly connected, Ethernet0/1
C
  20.0.0.1/32 is directly connected, Ethernet0/1
C
  21.0.0.0/8 [1/0] via 20.0.0.2
C
  30.0.0.0/8 [1/0] via 20.0.0.2
C
  162.14.0.0/16 [1/0] via 20.0.0.2
C
  172.19.0.0/16 [1/0] via 20.0.0.2
C
  192.168.168.0/24 [1/0] via 20.0.0.2
R1#sh ip int brief
Interface IP-Address OK? Method Status Prot
Gig0/0 10.0.0.1 YES manual up up
Ethernet0/1 20.0.0.1 YES manual up up
Ethernet0/2 unassigned YES TFTP down down
Ethernet0/3 unassigned YES TFTP down down
R1#
```

Assign IP to the PC



Ping From PC

The image displays three sequential screenshots of a network simulation environment, specifically PNETLab, showing a PC (Win-1) performing ping tests on various servers within a complex network topology. The topology includes multiple routers (R1, R2, R3, R4, R5, R6, R7, R8, R9, R10, R11, R12, R13, R14, R15, R16, R17, R18, R19, R20, R21, R22, R23, R24, R25, R26, R27, R28, R29, R30, R31, R32, R33, R34, R35, R36, R37, R38, R39, R40, R41, R42, R43, R44, R45, R46, R47, R48, R49, R50, R51, R52, R53, R54, R55, R56, R57, R58, R59, R60, R61, R62, R63, R64, R65, R66, R67, R68, R69, R70, R71, R72, R73, R74, R75, R76, R77, R78, R79, R80, R81, R82, R83, R84, R85, R86, R87, R88, R89, R90, R91, R92, R93, R94, R95, R96, R97, R98, R99, R100) and servers (S1, S2, S3, S4, S5, S6, S7, S8, S9, S10, S11, S12, S13, S14, S15, S16, S17, S18, S19, S20, S21, S22, S23, S24, S25, S26, S27, S28, S29, S30, S31, S32, S33, S34, S35, S36, S37, S38, S39, S40, S41, S42, S43, S44, S45, S46, S47, S48, S49, S50, S51, S52, S53, S54, S55, S56, S57, S58, S59, S60, S61, S62, S63, S64, S65, S66, S67, S68, S69, S70, S71, S72, S73, S74, S75, S76, S77, S78, S79, S80, S81, S82, S83, S84, S85, S86, S87, S88, S89, S90, S91, S92, S93, S94, S95, S96, S97, S98, S99, S100). The PC (Win-1) is connected to the network and is shown performing ping tests on various servers. The screenshots show the PC's command prompt with the following commands and output:

```
C:\Windows\system32\cmd.exe
C:\Documents and Settings\user>ipconfig

Ethernet adapter Local Area Connection:

   Connection-specific DNS Suffix  . : 
   IP Address. . . . . : 10.0.0.10
   Subnet Mask . . . . . : 255.255.255.0
   Default Gateway . . . . . : 10.0.0.1

C:\Documents and Settings\user>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

Reply from 10.0.0.10: bytes=32 time=0ms TTL=128
Reply from 10.0.0.10: bytes=32 time=0ms TTL=128
Reply from 10.0.0.10: bytes=32 time=0ms TTL=128
Reply from 10.0.0.10: bytes=32 time=0ms TTL=128

Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milliseconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Documents and Settings\user>ping 15.0.0.10

Pinging 15.0.0.10 with 32 bytes of data:

Reply from 15.0.0.10: bytes=32 time=0ms TTL=128
Reply from 15.0.0.10: bytes=32 time=0ms TTL=128
Reply from 15.0.0.10: bytes=32 time=0ms TTL=128
Reply from 15.0.0.10: bytes=32 time=0ms TTL=128

Ping statistics for 15.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milliseconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Documents and Settings\user>ping 162.14.0.10

Pinging 162.14.0.10 with 32 bytes of data:

Reply from 162.14.0.10: bytes=32 time=0ms TTL=128
Reply from 162.14.0.10: bytes=32 time=0ms TTL=128
Reply from 162.14.0.10: bytes=32 time=0ms TTL=128
Reply from 162.14.0.10: bytes=32 time=0ms TTL=128

Ping statistics for 162.14.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milliseconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Documents and Settings\user>
```

The screenshots show the PC's command prompt with the following commands and output:

```
C:\Windows\system32\cmd.exe
C:\Documents and Settings\user>ipconfig

Ethernet adapter Local Area Connection:

   Connection-specific DNS Suffix  . : 
   IP Address. . . . . : 10.0.0.10
   Subnet Mask . . . . . : 255.255.255.0
   Default Gateway . . . . . : 10.0.0.1

C:\Documents and Settings\user>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

Reply from 10.0.0.10: bytes=32 time=0ms TTL=128
Reply from 10.0.0.10: bytes=32 time=0ms TTL=128
Reply from 10.0.0.10: bytes=32 time=0ms TTL=128
Reply from 10.0.0.10: bytes=32 time=0ms TTL=128

Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milliseconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Documents and Settings\user>ping 15.0.0.10

Pinging 15.0.0.10 with 32 bytes of data:

Reply from 15.0.0.10: bytes=32 time=0ms TTL=128
Reply from 15.0.0.10: bytes=32 time=0ms TTL=128
Reply from 15.0.0.10: bytes=32 time=0ms TTL=128
Reply from 15.0.0.10: bytes=32 time=0ms TTL=128

Ping statistics for 15.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milliseconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Documents and Settings\user>ping 162.14.0.10

Pinging 162.14.0.10 with 32 bytes of data:

Reply from 162.14.0.10: bytes=32 time=0ms TTL=128
Reply from 162.14.0.10: bytes=32 time=0ms TTL=128
Reply from 162.14.0.10: bytes=32 time=0ms TTL=128
Reply from 162.14.0.10: bytes=32 time=0ms TTL=128

Ping statistics for 162.14.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milliseconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Documents and Settings\user>
```

The screenshots show the PC's command prompt with the following commands and output:

```
C:\Windows\system32\cmd.exe
C:\Documents and Settings\user>ipconfig

Ethernet adapter Local Area Connection:

   Connection-specific DNS Suffix  . : 
   IP Address. . . . . : 10.0.0.10
   Subnet Mask . . . . . : 255.255.255.0
   Default Gateway . . . . . : 10.0.0.1

C:\Documents and Settings\user>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

Reply from 10.0.0.10: bytes=32 time=0ms TTL=128
Reply from 10.0.0.10: bytes=32 time=0ms TTL=128
Reply from 10.0.0.10: bytes=32 time=0ms TTL=128
Reply from 10.0.0.10: bytes=32 time=0ms TTL=128

Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milliseconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Documents and Settings\user>ping 15.0.0.10

Pinging 15.0.0.10 with 32 bytes of data:

Reply from 15.0.0.10: bytes=32 time=0ms TTL=128
Reply from 15.0.0.10: bytes=32 time=0ms TTL=128
Reply from 15.0.0.10: bytes=32 time=0ms TTL=128
Reply from 15.0.0.10: bytes=32 time=0ms TTL=128

Ping statistics for 15.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milliseconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Documents and Settings\user>ping 162.14.0.10

Pinging 162.14.0.10 with 32 bytes of data:

Reply from 162.14.0.10: bytes=32 time=0ms TTL=128
Reply from 162.14.0.10: bytes=32 time=0ms TTL=128
Reply from 162.14.0.10: bytes=32 time=0ms TTL=128
Reply from 162.14.0.10: bytes=32 time=0ms TTL=128

Ping statistics for 162.14.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milliseconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Documents and Settings\user>
```

Telnet & SSH

```
R1#
R1>conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ip domain-name ak.com
R1(config)#crypto key generate rsa modulus 2048
The name for the keys will be: R1.ak.com

% The key modulus size is 2048 bits
% Generating 2048 bit RSA keys, keys will be non-exportable...
[OK] (elapsed time was 4 seconds)

R1(config)#
*Jul 19 15:44:15.490: %SSH-5-BNABLED: SSH 1.9% has been enabled
R1(config)#ip ssh version 2
R1(config)#username ak privilege 15 secret ak
R1(config)#line vty 0 4
R1(config-line)#transport input ssh
R1(config-line)#login local
R1(config-line)#exit
R1(config)#
*Jul 19 15:46:33.797: %SYS-5-CONFIG_I: Configured from console by ak on vty0 (16
2.14.0.10)
R1(config)#
```

